

- AssemblyAI Voice Agent Hackathon

TempCheck

Voice-logged food safety — read back and proven. Every flagged temperature is spoken back and confirmed before it's saved, and every accuracy number in this deck was measured against AssemblyAI's real API, not estimated.

THE PROBLEM

Paper logs, wet gloves, one misheard digit

- Every US food establishment is required to log temperatures — most of it is still clipboards, checked with wet, gloved hands, hours after the fact.
- A 38°F reading logged as 48°F isn't a UX nitpick — it's a real food-safety failure, and it's invisible until an inspector (or an outbreak) finds it.
- The fix has to be hands-free (cooks are working) and has to catch the misheard digit *in the same breath*, not at the next audit.

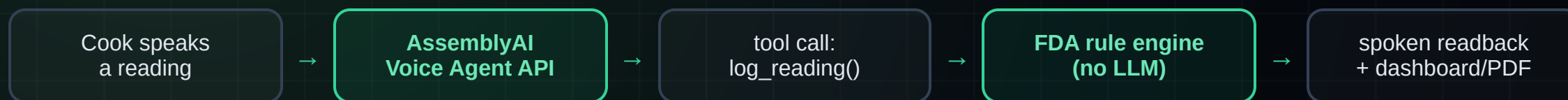
THE IDEA

Say it once. Hear it read back. It's logged.

- A cook calls out a reading mid-task — "walk-in cooler, thirty-eight," "chicken breast, one fifty-two."
- TempCheck logs it, checks it against real FDA limits, and speaks the exact number back before saving anything flagged.
- Out-of-range readings get a spoken corrective action, not a silent red dot nobody sees until the shift ends.

HOW IT WORKS

Cook → Voice Agent → deterministic verdict → readback



The Voice Agent handles STT, turn-taking, and tool-calling. It never decides whether a reading is safe — that's a separate, unit-tested, deterministic step.

The safety verdict is never a language model's guess

- The model's only job is extraction: location/item, number, unit. A plain, unit-tested comparison against FDA Food Code limits decides safe / amber / red.
- Cold holding $\leq 41^{\circ}\text{F}$ · hot holding $\geq 135^{\circ}\text{F}$ · poultry $\geq 165^{\circ}\text{F}$ · ground meat $\geq 155^{\circ}\text{F}$ · whole muscle / fish / eggs $\geq 145^{\circ}\text{F}$.
- 25 automated tests cover every category, every boundary, unit conversion, and implausible readings — zero hallucination risk in the verdict itself.

Two real integrations, not one API call

USE	API	WHAT IT DOES
Live cook-facing app	Voice Agent API	Real-time STT + turn detection + tool-calling + TTS, over a live WebSocket session
Accuracy proof harness	Async transcription API (v2)	270 real clips, real upload → transcript → poll, scored against ground truth

We didn't just call the API and hope — we built a real, reproducible test harness against it and published the actual result.

MEASURED ACCURACY — NO INVENTED NUMBERS

254 / 270 correct — 94.1%

96.7%

Clean audio (87/90)

98.9%

Moderate noise, ~12dB SNR
(89/90)

86.7%

Heavy kitchen noise, ~4dB SNR
(78/90)

Method disclosed *before* the result in `docs/accuracy.md`. The first real run scored 92.2% and caught three real bugs in our own number parser — fixed, re-tested against the live API again, and republished honestly.

LIVE DEMO

Dashboard, mid-shift

38°F

Walk-in cooler — safe

130°F

Steam table — amber

149°F

Ground beef — red, corrective
action prompted

Every entry keeps the cook's exact spoken words next to the verdict, and the whole shift exports as an inspector-ready HACCP PDF with one click.

Every kitchen already has to do this

- Temperature logging isn't optional — it's a health-code requirement at every US food establishment.
- TempCheck turns a paper chore into a real-time, hands-free, provably-checked habit, with the exported PDF already in the format an inspector expects.
- No new hardware: it runs in a browser tab a kitchen already has open.

ORIGINALITY

A narrow angle in a gallery of general-purpose agents.

Most submissions in this hackathon build general-purpose copilots, dispatch/fraud agents, or interview coaches. TempCheck is deliberately narrow: one real workflow, done with enough rigor — deterministic rules, mandatory readback, measured accuracy, real end-to-end verification — that it could plausibly go into a real kitchen tomorrow.

WHAT'S NEXT

Known limitations, stated plainly

- Cooling-curve time tracking (135°F → 70°F within 2h, then → 41°F within 4h) — only point-in-time readings are checked today.
- No persistent multi-shift storage yet — a shift's log lives in the browser tab and exports to PDF before ending.
- Phone/SIP deployment for kitchens that don't want a browser tab open.

LINKS

Try it yourself

- Live demo: tempcheck-vsh.vercel.app
- Code: github.com/pranuthi777/tempcheck-vs
- Measured accuracy & method: [docs/accuracy.md](#)